Department of Mathematics and Natural Sciences
Section - 06, Quiz - 01*, Spring 2025
Course Code: MAT110, Course Title: MATH I: Differential Calculus and Co-ordinate Geometry

Time: 20 minutes | Marks: 25

N.B. Answer all of the following questions. Make sure to show your steps and calculations as appropriate.

1.	Evaluate the limit, if it exists, for: $$a)\ \lim_{x\to\infty} \frac{x+x^3}{2-x^2+3x^3} \qquad b)\ \lim_{x\to 0} -\frac{\sqrt{9+x}-3}{x}$$	7 + 7
2.	Test the continuity and differentiability of the following function at $x=0$: $$f(x)=\begin{cases} \cos x & \text{if } x<0 \\ 0 & \text{if } x=0 \\ 1-x^2 & \text{if } x>0 \end{cases}$$ If the function is not continuous, state the type of discontinuity.	11

–Best of luck–